Contrastive Learning-Based Deep Embedded Clustering and the TCN-DMAttention Model for Traffic Congestion Prediction

Biao Zhang, Haoyu Zhang, *Member, IEEE*, Wenlin Li, Nisang Chen, and Xuchu Jiang

***Abstract*—With the increasing number of motor vehicles per capita, the problem of road traffic congestion is becoming increasingly prominent. Intelligent transportation systems are playing an increasingly important role in traffic management. Establishing a reasonable and effective traffic congestion prediction model is crucial for the implementation of intelligent transportation systems. This article proposes an end-to-end congestion prediction framework for unsupervised learning. First, a traffic data deep embedding clustering model based on contrast learning (CL-DEC) was constructed to discretize and label the congestion status of spatiotemporal rasterized GPS data. Then, the Time Convolutional Network (TCN) is used to extract features from the labeled data, and the Dependency Matrix Attention Mechanism (DMAttention) is combined to highlight important features for predicting traffic congestion. In this study, we found that DEC can effectively alleviate the false positive problem of CL, and the combination of the two improves discriminability and comparability, which helps to improve the accuracy of data annotation. In addition, TCN-DMAttention can balance the short-term fluctuations and long-term trends of traffic time series data. We compared it with 13 baseline models and conducted robustness analysis using 20 random number seeds. The experimental results indicate that the TCN-DMAttention model has better predictive performance and good stability than other comparative models do.**

***Index Terms*—Traffic congestion prediction, contrastive learning, deep embedded clustering, temporal convolutional network, attention mechanism.**

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Data is available on-line at https://github.com/1298242755/traffic-congestion-prediction

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I. Introduction

With he increase in per capita vehicle ownership and the acceleration of urbanization, urban traffic demand and pressure continue to increase, and traffic congestion is becoming increasingly severe. Frequent traffic congestion is a constraint on human health and economic development in multiple ways. Studies have shown that traffic congestion causes air pollution, increases time costs, mental stress and other hazards [1] and is one of the causes of more than 60% of traffic accidents worldwide [2]. Intelligent transportation has become a current trend. Accurate prediction of traffic congestion and improvement of traffic network efficiency are important topics in the research field of intelligent transportation systems, which will help optimize the allocation of road traffic resources and assist the government in further planning and expanding the urban road network [3].

In early decades, owing to the lack of data availability, classical statistical methods were limited to predicting the traffic parameters of individual roads and local networks [4]. At present, traffic congestion data come from two main sources: sensor and fixed probe data and vehicle GPS data [5]. The large amount of data and abundant parameters provide a solid foundation for the development of more global traffic congestion predictions. However, these raw data are usually not structured data but rather time series data with characteristics such as high frequency and dimensionality, and traditional statistical models are no longer applicable.

Rapid advances in machine learning and deep learning have provided new solutions to this problem. These methods usually need to perform supervised learning on the data that have completed the congestion degree annotation, which is highly dependent on the data annotation. (1) Network training can be conducted directly on the basis of existing datasets with relevant labels such as PeMS [6], UCSD [7] and data collected from the Amap and Didi platforms. (2) First-hand data that can be directly obtained, such as sensor/fixed probe data and vehicle GPS data, often lack relevant labels for congestion degree. For such datasets, the classification strategy is superior to the regression strategy in real situations for predicting the congestion state [8]. Therefore, the existing methods generally first map traffic flow features such as travel time and speed to the degree of congestion via discretization or use a simple autoencoder to extract features and label the degree of traffic congestion combined with a classification model to continue classification prediction of time series data.

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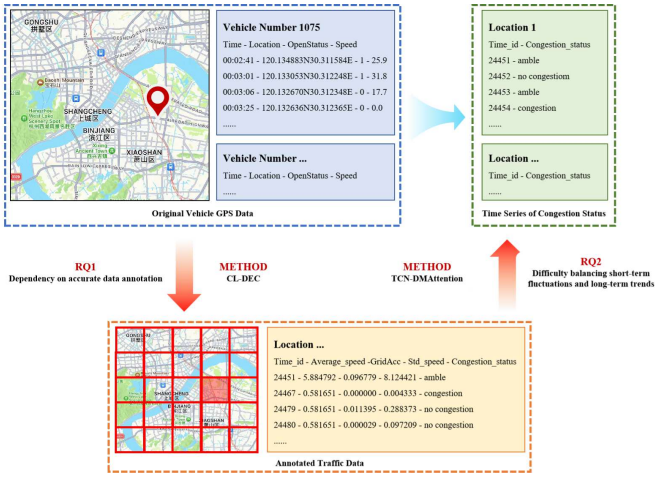

Fig. 1. Main research questions.

However, the existing methods have difficulty improving the efficiency and accuracy of traffic congestion prediction. In particular, the following research questions (RQ1, RQ2) need to be solved, as shown in Fig. 1:

RQ1: Dependency on accurate data annotation: (1) The annotated dataset is usually not the first-hand data generated in real time and is difficult to update with the generation of new data in practical applications because of strong limitations in online learning [9]. (2) Sensor/fixed probe data and GPS data reflect real-time changes in speed, position and other variables over time. The existing methods for marking the degree of congestion have difficulty effectively extracting the time series invariance features of traffic flow datasets. The labeling is rough, and the reliability of the prediction results is low.

RQ2: Difficulty balancing short-term fluctuations and long-term trends: Existing methods have a good ability to extract short-term fluctuations, but long-term trends (such as slow driving gradually evolving into congestion) still play an important role in congestion prediction, which is often overlooked.

To overcome the above limitations, this paper combines previous research results and proposes an end-to-end traffic congestion prediction framework for unsupervised learning. First, a traffic data deep embedding clustering model (CL-DEC) based on comparative learning was constructed to label the congestion state of spatiotemporal rasterized GPS data. Second, the temporal convolutional network dependence matrix attention model (TCN-DMAttention) is used to predict the congestion state. The main contributions of this paper are summarized as follows:

(1) An explicable deep embedded clustering model (CL-DEC) based on contrast learning is proposed, which uses a KL divergence-based loss function to optimize the encoder and effectively alleviates the "false positive" problem in comparative learning. It combines discrimination and comparability, significantly improving the accuracy of temporal feature labeling.
(2) A TCN that combines dependency matrix attention (TCN-DMAttention) has been proposed, in which the TCN can capture dependency relationships over a longer time span through dilated convolutions, and the DMAttention dynamically adjusts weights to avoid long-term trends being overwhelmed by short-term fluctuations.
(3) Through the effective combination of CL-DEC and TCN-DMAtattention, an end-to-end traffic congestion prediction model of "input of unmarked first-hand traffic data - output traffic congestion state" is constructed, making it possible to introduce an online learning framework.

In this work, the proposed model was trained and tested on the taxi GPS dataset of Hangzhou, China, and 13 mature baseline models and 20 random number seeds were set to test the prediction effect and robustness. The experimental results show that the proposed method has good stability and the highest values of $Precision_{weighted}$, $F1_{weighted}$, AUC and $accuracy$.

The remainder of the paper is organized as follows: Section II summarizes the relevant work and, more specifically, highlights the advantages and disadvantages of existing methods; Section III introduces the proposed method in detail. Section IV briefly analyzes the basic situation of the data and preprocessing methods and analyzes the interpretability of the data annotation methods. Section V describes the experimental process and details and analyzes the results in detail. Finally, Section VI presents the conclusions of this paper and directions for future research.

II. Related Work

Over the years, many scholars have proposed various traffic congestion prediction methods. In this section, a comprehensive review of the relevant literature is conducted from two perspectives: existing methods for traffic congestion prediction and feature extraction and annotation methods for time series data.

A. Traffic Congestion Prediction

1) Classical Statistical Model: Levin et al. used the ARIMA model to predict highway traffic flow and reported that ARIMA (0, 1, 1) yielded the highest statistical significance [10]. Kong et al. used floating cars as mobile sensors to detect large-scale urban traffic flow and used a new fuzzy comprehensive evaluation method to evaluate congestion. Moreover, the traffic prediction method based on the particle swarm optimization algorithm is adopted to calculate the traffic parameters, and finally, the traffic parameters are transformed into the congestion state recognized by citizens [11]. Cao et al. took traffic volume, speed and traffic density as the basic characteristics, followed Mamdani-type fuzzy logic reasoning, and obtained a description of the actual traffic state through mathematical and physical derivation [12].

With the development of intelligent transportation, current traffic big data are generally nonlinear and unstructured, and these statistical models are no longer applicable [13], [14].

2) Machine Learning and Deep Learning: In recent years, with the development of machine learning and deep learning research, an increasing number of scholars have applied advanced computational methods to analyze traffic congestion. Zhang et al. proposed a hierarchical fuzzy rule system based

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on a genetic optimization algorithm to establish a robust congestion prediction model [15]. Liu et al. leveraged the high robustness and practicality of the random forest algorithm to construct a traffic congestion state prediction model [16]. Fouladgar et al. proposed a method based on decentralized deep learning, which enables each node to accurately predict its own congestion state in real time on the basis of the congestion state of neighboring sites. Moreover, a regularized Euclidean loss function that favored high-congestion samples rather than low-congestion samples was introduced to avoid the influence of unbalanced samples. The coarse granularity issue is overcome by ignoring the spatial correlation between road segments [17]. Zhang et al. proposed a short-term traffic flow prediction model based on a convolutional neural network (CNN) deep learning framework and determined the optimal input data time delay and spatial data amount through the spatiotemporal feature selection algorithm [18]. Di et al. proposed the CPM-ConvLSTM model, which is based on a spatial matrix for the short-term prediction of the congestion level of each road section [19]. Chen et al. established the deep architecture of the cascade long short-term memory model for learning and predicting traffic conditions collected via web map services, and its effect was better than that of the multilayer perceptron model, decision tree model and support vector machine (SVM) model [20]. Huang et al. proposed a traffic congestion prediction method based on bus travel time combined with the use of long short-term memory (LSTM) to predict future bus congestion time and applied the congestion index and classification model to classify congestion into five categories to achieve effective prediction [21]. Ranjan et al. used online web services to obtain snapshots of traffic congestion maps and combined CNN, LSTM and transposed CNN to form a hybrid neural network architecture to predict the congestion level of the whole network through spatiotemporal information in the image [22]. Zhang et al. constructed a neural network model based on a deep autoencoder in which the encoder and decoder had symmetric layers and achieved the purpose of predicting traffic congestion by learning the temporal correlation of the traffic network [23]. Li et al. provided a comprehensive survey on Spatial-Temporal Graph Neural Networks (STGNNs), highlighting their effectiveness in capturing complex spatial–temporal dependencies in traffic data [24]. Building upon this, Kong et al. proposed a Self-Adaptive Spatial-Temporal Graph Attention Network that dynamically adjusts attention weights on the basis of both spatial and temporal factors [25]. Lim et al. introduced Temporal Fusion for interpretable multi-horizon forecasting and analyzed the interpretability of the traffic prediction model [26]. Chen et al. developed TrafficStream, a framework that combines graph neural networks with continual learning for streaming traffic flow forecasting [27]. Despite these advancements, challenges remain in balancing short-term fluctuations and long-term patterns while maintaining computational efficiency.

Manibardo et al. verified that addressing the problem as a classification task is superior to predicting the speed value and discretizing afterwards, and on the premise of paying attention to classification strategies, it also shows that online learning methods can significantly improve the prediction performance of traffic congestion, especially when dealing with concept drift and non-stationary data [8]. However, the above existing methods are highly dependent on the data label when performing supervised learning, and most of them use a dataset that has completed traffic congestion degree labeling for training. Moreover, owing to the lack of annotations, they are difficult to apply to the first-hand data generated by intelligent transportation systems (such as sensor/fixed probe data and GPS data), limiting the advantages of online learning frameworks.

To solve this problem, Wang J et al. used comprehensive records of vehicles equipped with GPS devices to measure the traffic conditions of each road segment via discretization of the average speed, established a mapping from the degree of road congestion to symbolic time series, and used information theory tools such as entropy and Fano inequality to measure the predictability of the traffic conditions of road segments [28]. Chiabaut et al. first clustered the observation days of historical datasets through principal component analysis and identified a representative consensus day in each cluster to predict the evolution of traffic congestion and corresponding travel time changes in a simple and interpretable way in real time. This method performs well over longer forecast time horizons [29]. Chen et al. defined congestion on the basis of average travel time and baseline travel time and proposed the PCNN method, which is based on a deep convolutional neural network. This method focuses on two key processes—time series folding and learning—for modeling periodic traffic data and can be used for short-term traffic congestion prediction [30].

Most of the existing studies focus on improving congestion prediction methods after data annotation, and few people pay attention to the importance of the process of determining the degree of congestion. Accurate data annotation results are the basic guarantee for optimized prediction methods, but the existing methods for congestion degree annotation have serious limitations: (1) Ignoring the time series invariance and other characteristics of traffic data, it is difficult to fully extract important features, and the annotation results may have large errors; (2) Compared with the direct mapping of congestion degree through speed and time, the method based on clustering to achieve discretization and generate pseudo-label can more fully synthesize multi-dimensional data features, but few studies in the field of traffic congestion prediction exist.

B. Feature Labeling of Time Series Data

Data annotation requires many resources, and excellent feature extraction methods can effectively optimize downstream tasks. Comparative learning can fully exploit temporal correlation and dynamic changes to optimize feature representation, which has great advantages when applied to time series data. Ren et al. proposed RankSCL, a rank-supervised contrast learning framework, which applies a new rank loss algorithm to enable encoders to extract fine-grained information of the same class and generate clear boundaries between different classes, thereby enhancing the original data and selecting

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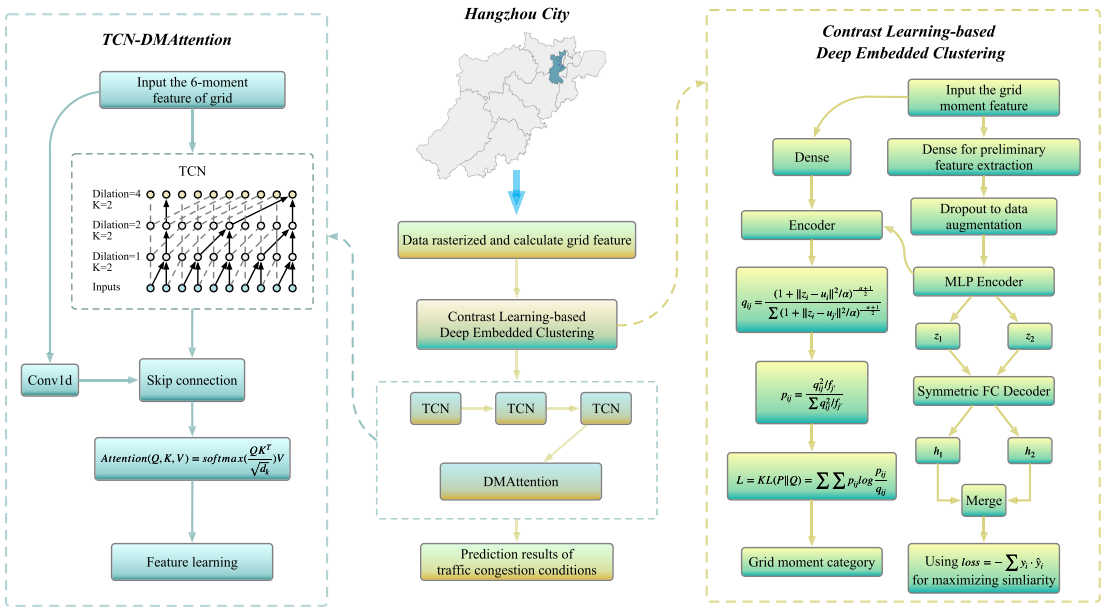

Fig. 2. Framework of CL-DEC and TCN-DMAttention.

positive and negative pairs of more information in the embedded space in a targeted way [31]. Luo et al. established a framework of principles for selecting enhancements on the basis of dataset features (such as trends and seasonality) and summarized the generable correlation between time series features and enhancement efficiency [32].

More importantly, contrast learning works well in learning effective representations of time series data with limited labels. Emadeldeen et al. proposed a novel time series representation learning framework through time and context contrast (TS-TCC), which reduces the need for time series data labeling by learning representations from unlabeled data through contrast learning and reasonable self-supervised and semi-supervised methods [33]. Cai et al. proposed SLOTS, an end-to-end semi-supervised learning time classification model that receives labeled datasets consisting of many unlabeled samples and a small number of labeled samples and maps them to the embedded space through an encoder [34]. EskandariNasab et al. adopted a new preprocessing process, including missing value input, normalization, balanced sampling, near-decision boundary sample removal and feature selection, and combined contrast learning with a GRU regression model to develop a new classifier ContReg, which improved the prediction performance of solar flares [35].

The time series feature extraction and data annotation methods based on contrast learning have significant advantages and wide application potential in processing unlabeled data and enhancing strategy selection. This paper further explores its applicability to traffic time series data.

III. Methodology

In this section, a new traffic congestion prediction framework is constructed via Contrast Learning (CL), Deep Embedded Clustering (DEC), and TCN-DMAttention modules.

A. Overview

The supervised learning model based on machine learning and deep learning has achieved good results in the field of traffic congestion prediction. However, for traffic data that have not yet been annotated, inadequate feature extraction and data annotation methods may lead to a large difference between the trained model and the expected effect in practical applications. To overcome this limitation, this paper proposes an end-to-end traffic congestion prediction framework that implements unsupervised learning. This framework starts with spatiotemporal rasterized GPS data and constructs a traffic data deep embedding clustering model (CL-DEC) based on contrast learning to discretize the congestion state. Moreover, to strike a balance between short-term fluctuations and long-term trends in learning history, we design a time series convolutional network dependence matrix attention mechanism model (TCN-DMAttention) to achieve real-time prediction of the congestion state, building upon the latest developments in the field as outlined by Wu et al. and Yin et al. in their comprehensive surveys on traffic prediction methods [36], [37]. The overall structure of the model is shown in Fig. 2.

In particular, (1) we fully consider the correlation between time and space in traffic problems, divide the road space into rectangular areas (grids) of equal size on the basis of vehicle GPS data, and calculate the driving data features within each grid, greatly simplifying the complexity of the problem. (2) To improve the accuracy of discretized labeling of the traffic congestion status, we construct CL-DEC, which divides the status of each grid into three categories (congestion, slow-moving, and unobstructed). Among them, the contrastive learning mechanism can optimize feature representation by pulling positive samples closer and pushing negative samples further away, resulting in a uniform distribution of the final samples in the latent space and enhancing the discriminability of the model. However, for samples that are similar in category

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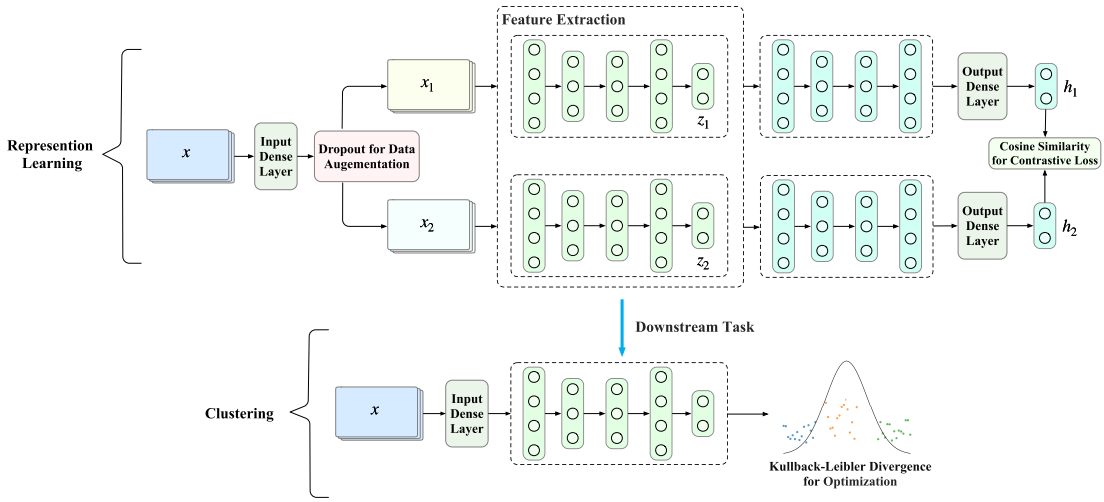

Fig. 3. Framework of CL-DEC.

to the target sample but have not been correctly identified, the model may mistakenly treat them as negative samples for the "push away" operation, resulting in unclear representation of the embedding space and affecting subsequent classification tasks. Deep embedding clustering minimizes the loss function based on KL divergence and guides the encoder to learn a more ordered and structured embedding space by pushing the probability distribution of samples in the latent space to be consistent with the target distribution, avoiding the problem of "false positives" caused by excessive optimization of contrastive loss and enhancing comparability at the same time. (3) In particular, to address the challenge of balancing short-term fluctuations and long-term trends, our TCN-DMAttention model employs complementary mechanisms to map the average speed, acceleration, speed standard deviation, congestion status, and other features of the first 6 moments to the next congestion state. The TCN component uses hierarchical dilated convolutions with varying dilation rates ($d = 1, 2, 4$) to capture traffic patterns simultaneously at multiple time scales: immediate changes (short-term), gradual congestion formation (medium-term), and recurring daily patterns (long-term). This multiscale feature extraction creates a comprehensive temporal representation of traffic dynamics. Moreover, the DMAttention mechanism dynamically weighs the importance of these temporal features on the basis of their relevance to the current prediction context. This adaptive temporal focus allows the model to emphasize short-term patterns during rapidly changing traffic conditions while giving appropriate weights to long-term trends during stable periods. The synergistic integration of these components enables our model to excel at identifying transition patterns—how short-term fluctuations evolve into long-term trends (e.g., how a series of slowdowns eventually leads to full congestion)—providing more accurate and robust predictions than approaches that overemphasize either immediate dynamics or longer-term evolution patterns.

B. CL-DEC

Comparative learning methods learn the similarities and differences between traffic time series data by comparing positive and negative samples, effectively capturing their temporal dependencies and dynamic changes, and extracting more discriminative features. These features can be effectively used for downstream tasks, such as clustering and labeling of traffic states. In clustering tasks, Deep Embedding Clustering (DEC) uses the Student t-SNE distribution as a measure of the similarity between embedded nodes and cluster centers to avoid poor clustering results caused by the high dimensionality of extracted data features. This refers to the probability that a grid sample i belongs to cluster center j, as shown in Eq. (1).

$$q_{ij} = \frac{\left(1 + \|z_i - u_i\|^2 / \alpha\right)^{-\frac{\alpha+1}{2}}}{\sum_{j'} \left(1 + \left\|z_i - u_{j'}\right\|^2 / \alpha\right)^{-\frac{\alpha+1}{2}}}. \tag{1}$$

where z_i is a high-quality embedding space vector obtained by comparing and learning feature representations of traffic time series, u_i is a clustering center for three types of traffic states, α is the degree of freedom of the t-distribution, which is generally set to 1, and the calculated q_{ij} is the probability that sample i is assigned to cluster center j. On the basis of the calculated p_{ij}, DEC constructs the target distribution by squaring the soft-assigned probability q_{ij}, as shown in Eq. (2).

$$p_{ij} = \frac{q_{ij}^2 / f_{j'}}{\sum_{j'} q_{ij}^2 / f_{j'}} \tag{2}$$

where $f_j = \sum_i q_{ij}$ represents the soft group frequency. The target distribution strengthens the prediction, improves confidence, regulates the loss of the cluster center, and prevents categories with large sample numbers from distorting the feature space. The distance between the two distributions mentioned above can be measured via the KL divergence, as shown in Eq. (3).

$$L = KL(P\|Q) = \sum_i \sum_j p_{ij} \log \frac{p_{ij}}{q_{ij}}. \tag{3}$$

The structure of the deep embedding clustering model (CL-DEC) based on contrastive learning is shown in Fig. 3. To establish the network structure for comparative learning,

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we first use a fully connected layer for preliminary feature representation of the input traffic time series x. We then apply data augmentation techniques, specifically dropout, to this representation to generate two different versions of the input, which we denote as x_1 and x_2. These augmented versions, x_1 and x_2, are considered positive sample pairs in the context of contrastive learning. This means that x_1 and x_2 are treated as different views of the same underlying data point, and the model is trained to recognize their similarity despite the augmentation-induced differences. In the selection of data enhancement methods, referring to the experimental results of Gao T et al. in simple contrastive learning of sentence embeddings, this paper uses dropout to perform data augmentation on sequence data [38].

In this work, a multilayer perceptron (MLP) is used to construct an encoder to learn the representations of x_1 and x_2 to obtain the feature vectors z_1 and z_2, respectively; then, the symmetric fully connected layer structure is used as the decoder to obtain the output vectors h_1 and h_2. The processed positive samples and negative samples are merged, the similarity matrix m_1 of each sample is calculated via cosine similarity, and the real matrix m_2 is constructed as shown in Eq. (4), where (h_1, h_2) represents a positive sample pair derived from the same input, while pairs involving l_1 and l_2 (such as (h_1, l_1), (h_2, l_2), etc.) represent comparisons with samples from different inputs. The matrix m1 captures all pairwise comparisons, including self-comparisons (diagonal elements) and cross-comparisons. The loss function is calculated on the basis of m_1 and m_2, as shown in Eq. (5).

$$m_1 = \begin{bmatrix} q_{(h_1,h_1)} & q_{(h_1,h_2)} & q_{(h_1,l_1)} & q_{(h_1,l_2)} \\ q_{(h_2,h_1)} & q_{(h_2,h_2)} & q_{(h_2,l_1)} & q_{(h_2,l_2)} \\ q_{(l_1,h_1)} & q_{(l_1,h_2)} & q_{(l_1,l_1)} & q_{(l_1,l_2)} \\ q_{(l_2,h_1)} & q_{(l_2,h_2)} & q_{(l_2,l_1)} & q_{(l_2,l_2)} \end{bmatrix}, \tag{4}$$

$$m_2 = \begin{bmatrix} 0 & 1 & 0 & 0 \\ 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 \\ 0 & 0 & 1 & 0 \end{bmatrix}.$$

$$\text{loss} = -\sum_i y_i \cdot \log \widehat{y}_l \tag{5}$$

where y_i and $\widehat{y}_i$ are the corresponding elements of the matrices m_2 and m_1, respectively.

C. TCN-DMAttention

The TCN-DMAttention module combines the temporal feature extraction capabilities of TCNs with a dependency matrix attention mechanism to balance short-term fluctuations and long-term trends in traffic data. The structure of TCN-DMAttention is shown in Fig. 4.

1) TCN: A TCN is composed of causal dilated convolution and residual blocks. Compared with recurrent neural networks (RNNs), such as long short-term memory (LSTM) networks and gated recurrent units (GRUs), this method can effectively avoid the problems of gradient disappearance and explosion in time series feature extraction. When the h-layer network at time t is extracted via causal convolution, the TCN uses only the characteristics of the $h-1$ layer at time t and the moments before it. In this way, time constraints are applied to the convolution area, which is more in line with the logical properties of the time series. The calculation is shown in Eq. (6).

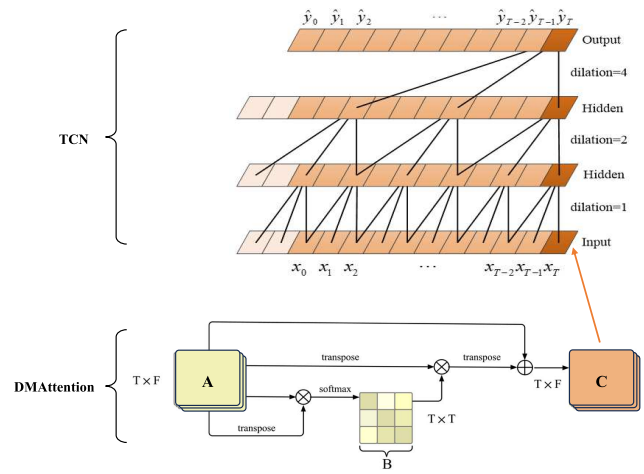

Fig. 4. Structure of the TCN-DMAttention.

$$y_{h,t} = \sum_{i=0}^{k} f_i \left(y_{h-1,t-2id}\right). \tag{6}$$

where t represents the current time step and $t-2id$ represents a time step that is 2 units of time (e.g., 2 hours, 2 days, etc.) before the current time. This allows the model to consider both recent and slightly older historical data in its predictions. $y_{h,t}$ is the sequence value of the h layer t time in the network, that is, the traffic characteristics of the road section. f_i, k, and d represent the number of filters in the network, the size of the convolution kernel and the convolution expansion rate, respectively. The residual connection in the TCN enables the network to transfer information across layers and prevents the network from distorting during the learning process.

2) DMAttention: DMAttention is a dependency matrix attention mechanism that is based on cosine similarity. In Fig. 4, DMAttention performs an inner product operation on the feature vectors of each moment and normalizes them by softmax to obtain the weight matrix reflecting the dependency relationship of each moment, as shown in Eq. (7).

$$x_{ij} = \frac{\exp\left(A_i^T \cdot A_j\right)}{\sum_i \exp\left(A_i^T \cdot A_j\right)}. \tag{7}$$

In Eq. (7), x_{ij} represents the dependence between time i and time j, and the weight reflects the similarity of feature vectors at two moments. If the eigenvectors of two moments have high cosine similarity, their corresponding weights will also be relatively high. The network matrix multiplies and transposes A^T and B to obtain the weighted sum value of the feature at each time, and finally, it is multiplied by the trainable scale coefficient γ and residually connected to the original feature to obtain the output C, as shown in Eq. (8).

$$C = \gamma \left(A^T \cdot B\right)^T + A. \tag{8}$$

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DMAttention can assign higher weights to feature vectors with higher similarity without being affected by time intervals. Unlike other attention mechanisms, such as time self-attention, which focuses on learning variable parameters, DMAttention models temporal correlations through "non parametric raw feature inner product similarity", emphasizing physical interpretability and weight. Therefore, it has higher efficiency and interpretability in scenarios where feature similarity can directly reflect dependency relationships, such as traffic temporal data prediction.

D. End-to-End Framework Integration

Our end-to-end framework seamlessly integrates the CL-DEC and TCN-DMAttention components to form a complete pipeline from raw GPS data to congestion prediction. The process begins with spatiotemporal rasterization of raw GPS data to create uniform grid-based representations, which are then fed into the CL-DEC module for unsupervised labeling. This module leverages contrastive learning to optimize feature representations and cluster the feature space into distinct congestion states (congested, slow-moving, and unobstructed), effectively assigning congestion labels to each grid cell at each time step without requiring prelabeled data. These labeled time series then serve as inputs for the TCN-DMAttention module, which extracts temporal features via dilated causal convolutions and applies the dependency matrix attention mechanism to balance short-term fluctuations with long-term trends, ultimately generating accurate predictions of future congestion states. This integration creates a highly adaptable framework suitable for new environments and online learning scenarios, while its modular design permits independent optimization of each component to enhance overall performance.

IV. Data Annotations

A. Dataset

At present, with the development and application of intelligent transportation, traffic data mainly come from sensors, fixed probes and vehicle GPSs. In practice, compared with the data of sensors and fixed probes, vehicle GPS data have the following advantages in the task of real-time traffic congestion prediction: (1) Wide coverage of data: fixed probes and sensors can usually cover only a specific road section or intersection, whereas vehicle GPS data can cover a wider area. (2) Strong real-time: Although fixed probes and sensors can collect data in real time, there may be a certain delay in data processing and transmission, and vehicle GPS data can be uploaded to the cloud in real time. The method proposed in this paper can predict and update road condition information in a timely and efficient manner. (3) Data diversity: Vehicle GPS data contain not only location information but also speed, driving direction and other diverse information, providing a richer data basis for traffic congestion prediction.

Therefore, the experimental data used in this paper are from taxi GPS positioning data of Hangzhou City on the data platform (http://tecdat.cn) from 0:00 to 24:00 on October 24, 2019, totaling 38484414 pieces of data. The vehicle information intervals in the data are variable, ranging from 4 to 30 seconds. The data contain six features, namely, the vehicle number, time, longitude, latitude, passenger carrying status, and speed. The first five data points are shown in TABLE I.

TABLE I
Taxi GPS Data

Number	VehicleNum	Stime	Lng	Lat	OpenStatus	Speed
1	1075	00:02:41	120.134883	30.311584	1	25.9
2	1075	00:03:01	120.133053	30.312248	1	31.8
3	1075	00:03:06	120.132670	30.312348	0	17.7
4	1075	00:03:06	120.132670	30.312348	0	17.7
5	1075	00:03:25	120.132636	30.312365	0	0.0

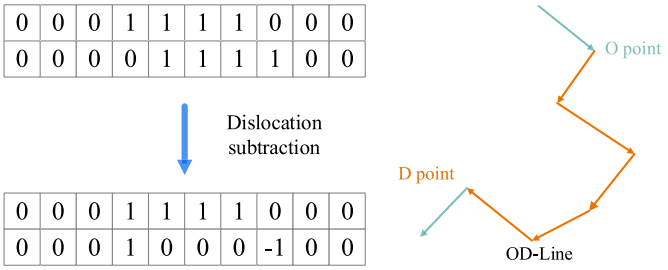

Fig. 5. OD-Line extraction.

Location information such as latitude and longitude in the original data cannot be used directly, and problems such as missing values, outliers and repeated records occur at the same time in the data; these problems need to be processed by data preprocessing for format conversion and outlier processing.

B. Data Preprocessing

GPS data are very large and prone to outliers. Reasonable data preprocessing can increase the accuracy of the clustering results and the prediction results. Moreover, taxi data cannot fully represent the traffic conditions of a city, but these data can be used to roughly describe traffic conditions by selecting data-intensive areas. In data preprocessing, the origin-destination line (OD-Line) and region constraints are used to select intensive travel areas and exclude outliers, and spatiotemporal rasterization is used to serialize the data to calculate various features. Finally, this paper uses CL-DEC to cluster grid traffic congestion.

1) Spatial Dimension: (1) Map rasterization

In this work, the regions within the latitude and longitude of Hangzhou city are divided into 400×400 grids of the same shape and size. Each grid is used in this paper as the smallest spatial unit to analyze the state of road congestion.

(2) Outlier handling and OD-Line extraction

In this paper, before OD-Line extraction, the data whose latitude and longitude are not within the scope of Hangzhou city, the data whose passenger carrying state is discontinuous, and the data whose recording time interval is 0 are deleted.

To eliminate interference such as autonomous parking during non-operation, the OD-Line was used to obtain the driving track of taxi passengers and the operation of the vehicle to assist in regional selection and outlier recognition. Specifically, the dislocation subtraction of the passenger carrying state of the vehicle is carried out to obtain the O and D points of the

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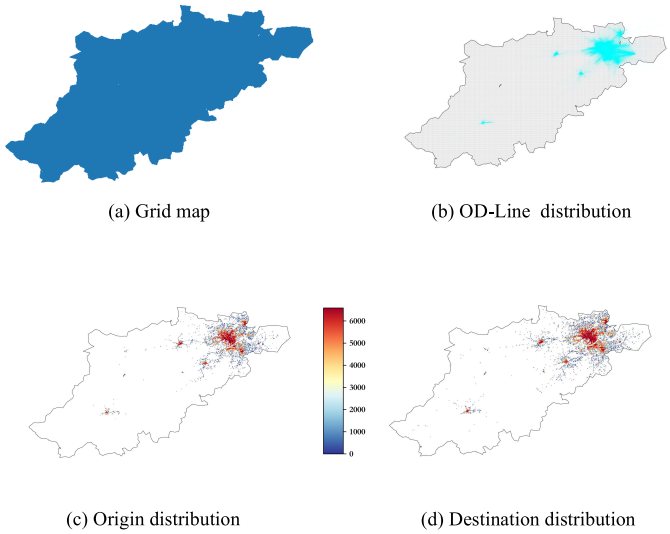

(a) Grid map (b) OD-Line distribution

(c) Origin distribution (d) Destination distribution

Fig. 6. OD-Line of Hangzhou city.

TABLE II
FEATURES AFTER SPATIOTEMPORAL RASTERIZATION

Number	rowid	colid	time_id	aveSpeed	gridAcc	speed_std	stopNum
1	0	22	24451	5.884792	−0.096779	8.124421	0.0
2	0	22	24467	0.581651	−0.000000	0.004333	3.0
3	0	22	24479	0.581651	−0.011395	0.288373	0.0
4	0	22	24480	0.581651	−0.000029	0.097209	0.0
5	0	22	24481	0.581651	−0.058463	0.130610	1.0

vehicle driving trajectory, and the connection between the two points is the OD-Line of the vehicle, as shown in Fig. 5.

(3) Region selection

To describe the running conditions of taxis and extract the features of time series, the vector trajectory is converted into grid data corresponding to the rasterized map. The distribution diagrams of the OD-Line, O point and D point are shown in Fig. 6.

As shown in Fig. 6(b), the taxi distribution in Hangzhou has obvious spatial differences and is concentrated in Xihu District in the northeast region of the city. As shown in Fig. 6(c) and Fig. 6(d), the distributions of taxi OD lines in Hangzhou are roughly the same. In summary, to increase the credibility of traffic prediction, this paper chooses Xihu District, where the OD line is concentrated, as the analysis region construction model.

2) Time Dimension: (1) Time window division

To reflect the time sequence of the data and meet the input format of the neural network model, this paper converts the time stamp of the rasterized GPS data and divides the time window. A total of 1200 time windows of equal length are divided; that is, the time interval of the structured data is set to 20 minutes.

(2) Feature calculation

On the basis of spatial rasterization and time segmentation, the average speed, average acceleration and standard deviation of each grid in the unit time window are calculated. Moreover, the average number of stops of the grid is calculated with a speed of 0 as the judgment basis for vehicle parking.

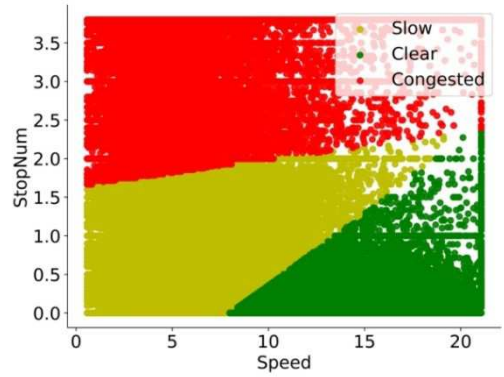

Fig. 7. Clustering results for traffic congestion conditions.

TABLE III
PARAMETER SETTINGS FOR CL-DEC

Item	Parameter type	Value
Contrastive Learning	Batch Size	20000
	Training Epochs	100
	Dropout Rate	0.1
	Loss Function	MSE
	Optimizer	ADAM
	Validation Split	0.25
Encoder	Input Dimension	2
	Layer Structure	$8 \to 32 \to 64 \to 128 \to 8$
	Activation Function	RELU
Decoder	Layer Structure	$128 \to 64 \to 32 \to 8 \to 8 \to 2$
	Activation Function	RELU
	Output Dimension	2
Clustering	Number of Clusters	3
	K-means Initialization	$n_init = 200$
Deep Clustering Training	Optimizer	SGD(0.01, 0.9)
	Loss Function	KLD
	Max Iterations	1000
	Update Interval	140
	Batch Size	128
	Early Stop Tolerance	0.01

After the original data were processed according to the above process, a total of 1,320,729 pieces of spatiotemporal rasterized data were obtained, as shown in TABLE II.

C. Annotation of Congestion Status

1) CL-DEC Clustering and Data Annotation: Considering interpretability and actual traffic conditions, this paper selects the average number of stops (more frequent in congestion) and average speed (slower in congestion) of a grid in the unit time window as the clustering features. The number of cluster centers of the cluster is 3, indicating that the traffic status of the grid at that moment is congested, slow and smooth.

The preprocessed data were brought into CL-DEC for unsupervised classification. The main parameter settings of CL-DEC are shown in TABLE III. The clustering results are shown in Fig. 7. For a grid at a certain time, the category of its cluster is used as the corresponding congestion state label in this paper.

2) Interpretation and Analysis of Results: The deep embedding model based on CL has a good clustering effect on traffic data and is interpretable: (1) When the average grid speed of vehicles is low, the average number of vehicles stopping is low, the average number of vehicles stopping is low, and

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TABLE IV
CONFUSION MATRIX

Confusion matrix		Predicted category Positive	Negative
Real category	Positive	TP	FN
	Negative	FP	TN

the average number of vehicles stopping is congested. When the average speed of vehicles in the grid is fast, the average number of vehicle stops is low, which means smooth, and the average number of vehicle stops is high, which means slow or congested. (2) When the average number of stops is small, the average speed of the vehicle is slow, and the average speed of the vehicle is smooth. When the average number of stops is large, the traffic pattern is congested. The above analysis and results show that it is effective to use CL-DEC clustering results for discrete labeling of traffic data.

V. EXPERIMENTAL ANALYSIS

A. Experimental Design

This paper divided the data into a training set, verification set and test set at a ratio of 6:2:2 in chronological order. After comprehensive consideration of the calculation cost and prediction accuracy, the number of input time windows is set to 6, and the input features are the average speed, acceleration, speed standard deviation and congestion conditions corresponding to the time window used to predict the traffic congestion status in the next time window.

1) Evaluation Metrics: The evaluation metrics used in this paper are based on the confusion matrix, as shown in TABLE IV.

In this work, the weighted precision ($Precision_{weighted}$), weighted F1 value ($F1_{weighted}$), *AUC* value and *accuracy* suitable for multiclassification tasks are calculated, where the weights are calculated according to the class sample size, as shown in Eq. (9).

$$w_i = \frac{c_i}{\sum_{j=1}^{n} c_j} \tag{9}$$

$Precision_{weighted}$ can be calculated through Eqs. (10) and (11).

$$Precision_i = \frac{TP_i}{TP_i + FP_i} \tag{10}$$

$$Precision_{weighted} = \sum_{i=1}^{n} w_i \cdot Precision_i. \tag{11}$$

$F1_{weighted}$ needs to be calculated by calculating the weighted recall ratio ($Recall_{weighted}$), as shown in Eqs. (12)–(15).

$$Recall_i = \frac{TP_i}{TP_i + FN_i}. \tag{12}$$

$$Recall_{weighted} = \sum_{i}^{n} wi \cdot Recall_i. \tag{13}$$

$$F1_i = \frac{2Precision_i \cdot Recall_i}{Precision^{+}Recall_i}. \tag{14}$$

TABLE V
HYPERPARAMETER SETTINGS FOR TCN-DMATTENTION

Item	Hyperparameter type	Value
Dilated causal conv layer 1	Dilation rate	1
	Convolution kernels	16
	Kernel size	3
Dilated causal conv layer 2	Dilation rate	2
	Convolution kernels	16
	Kernel size	3
Dilated causal conv layer 3	Dilation rate	4
	Convolution kernels	8
	Kernel size	3
DMAttention	-	-
Optimizer	Adam	Default

Fig. 8. Heatmap of the traffic congestion prediction results.

$$F1_{weighted} = \sum_{i}^{n} w_i \cdot F1_i. \tag{15}$$

AUC is calculated via Eq. (16).

$$AUC = P\left(P_{positive\ sample} > P_{negtive\ sample}\right) \tag{16}$$

These four metrics evaluate the model in different aspects, and an excellent model should obtain a large value for these four metrics.

2) Main Hyperparameter Settings: In this work, the data obtained from preprocessing are min–max normalized and input into the TCN-DMAttention model to predict the traffic congestion status. The main hyper-parameters of the TCN-DMAttention model constructed in this paper are shown in TABLE V.

Considering that the category imbalance problem of multiclassification tasks interferes with model fitting, this paper uses the focal loss function proposed by Lin et al. [39], which improves the fitting ability of the model for class imbalance tasks by giving more weight to the samples that are difficult to distinguish.

B. Analysis of Model Prediction Performance

1) Accuracy: We compare the congestion state results predicted by the model with reality, and the results are shown

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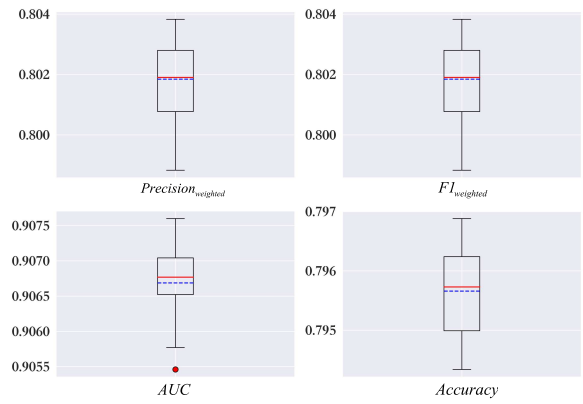

Fig. 9. Prediction results for 20 random number seeds.

TABLE VI
MEAN AND VARIANCE OF INDICATORS UNDER 20 RANDOM NUMBER SEEDS

Indicator	$Precision_{weighted}$	$F1_{weighted}$	AUC	$Accuracy$
Mean	0.8018	0.8018	0.9067	0.7957
Variance	1.6789e-06	1.6789e-06	2.8897e-07	6.5979e-07

TABLE VII
RESULTS OF THE CONTRAST EXPERIMENTS

Type of model	Model	$Precision_{weighted}$	$F1_{weighted}$	AUC	$Accuracy$
Proposed model	**TCN-DMAttention**	**0.8023**	**0.7957**	**0.9067**	**0.7962**
Deep learning model	RNN [40]	0.7830*	0.7725*	0.8741*	0.7760*
	TCN [41]	0.7836*	0.7787*	0.8923	0.7794*
	Ddnet [42]	0.7867*	0.7800*	0.8921	0.7803*
	MLP [43]	0.7761*	0.7450**	0.8596**	0.7573**
	GRU [44]	0.7816*	0.7740*	0.8788**	0.7762*
	LSTM [45]	0.7883*	0.7848*	0.8911	0.7855*
	xLSTM [46]	0.8015	0.7531**	0.9047	0.7661*
Machine learning model	Decision tree [47]	0.7888*	0.7785*	0.8796*	0.7789*
	AdaBoost [48]	0.7870*	0.7843	0.8921	0.7845*
	Logistic regression [49]	0.7834*	0.7737*	0.8713*	0.7769*
	Random forest [50]	0.7808*	0.7785*	0.8832*	0.7788*
	KNN [51]	0.7567**	0.7540**	0.8441**	0.7542*
	SVM [52]	0.7833*	0.7713*	0.8594**	0.7751*

in Fig. 8. The TCN-DMAttention model constructed in this paper is better at predicting traffic congestion, but the attention given by the category imbalance does not affect the fitting and prediction of the model.

2) Robustness: To analyze the robustness of the model, this paper uses 20 random number seeds to fit the model and analyze the mean and variance of the fitting results of the model according to the characteristic that the fitting results of the neural network model are affected by parameter initialization. In this work, model fitting is carried out under 20 different random number seeds, four metrics are calculated, box plots are shown in Fig. 9, and the mean and variance are shown in Table VI.

According to the experimental results in TABLE VI, the TCN-DMAttention model proposed in this paper has good robustness in traffic congestion prediction, and the prediction results perform well and have small variance under different random number seeds.

C. Comparative Analysis With Baselines

1) Comparison Results: In terms of model analysis, (1) to analyze the accuracy of the prediction results of the TCN-DMAttention model, in this paper, six mature machine learning models and seven neural networks are used to compare the four metrics $Precision_{weighted}$, $F1_{weighted}$, AUC and $Accuracy$. (2) To analyze the validity of DMAttention, this paper compares TCN-DMAttention with the TCN model. (3) To analyze the effectiveness of the TCN structure in temporal feature extraction on traffic datasets, this paper compares the TCN with the MLP. Moreover, to verify the effectiveness of our proposed method in improving performance, we set 20 random seeds for each of the 13 baseline models, trained these models with input data, and finally used analysis of variance to test whether the difference between our method and the baseline model was significant on the basis of the test results. The results of the comparative experiment and analysis of variance are shown in Table VII.

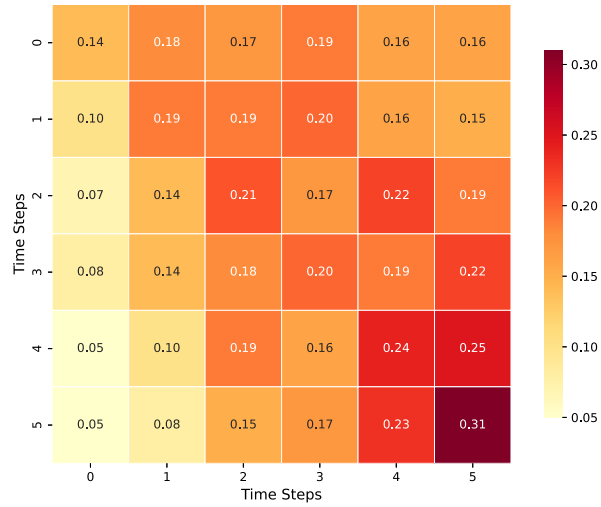

Fig. 10. Heatmap of attention.

According to the results of the comparative experiments in TABLE VII, the following conclusions can be drawn: (1) TCN-DMAttention outperforms the other 12 comparison models in terms of four evaluation indicators and is mostly significant at the 10% level, demonstrating good predictive performance for traffic congestion. (2) By comparing TCN-DMAttention and TCN, we find that the DMA attention structure has a positive effect on improving the prediction accuracy of the model in the network designed in this paper. (3) A comparison between the TCN and MLP methods reveals that the TCN structure, which is based on causal unfolding convolution and residual connections, can effectively extract traffic data features.

2) Interpretation and Analysis of Weights: Fig. 10 presents the attention weight heatmap, offering crucial insights into our model's temporal focus and decision-making process. The heatmap reveals a sophisticated pattern of temporal dependencies, with attention weights gradually intensifying from earlier to more recent time steps. This is evidenced by the color gradient shifting from lighter shades in the upper-left corner (representing earlier time steps) to darker shades in the lower-right corner (representing more recent time steps), culminating in the highest attention weight of 0.31 at position

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TABLE VIII
ABLATION STUDY RESULTS

Model Variant	$Precision_{weight}$	$F1_{weight}$	AUC	$Accuracy$
Full model	**0.8023**	**0.7957**	**0.9067**	**0.7962**
Without CL-DEC	0.7845	0.7789	0.8912	0.7798
Without CL	0.7959	0.7894	0.8963	0.7902
Without DEC(namely, CC)	0.7903	0.7836	0.8941	0.7875
TCN only	0.7901	0.7836	0.8978	0.7842
DMAttention only	0.7758	0.7692	0.8845	0.7701

(5, 5). This pattern underscores the model's prioritization of recent historical data in making predictions, which aligns with the dynamic nature of traffic flows where current conditions are heavily influenced by the immediate past. However, the model does not entirely disregard older information, as evidenced by non-zero weights in earlier time steps (ranging from 0.05 to 0.14 at time step 0). This balanced approach allows the model to capture both short-term fluctuations and longer-term trends in traffic patterns. The pronounced diagonal pattern in the heatmap, with higher values along and near the diagonal, suggests that the model emphasizes the relationships between adjacent time steps, capturing the continuity and evolution of traffic states over time. Interestingly, the asymmetry in the heatmap, with generally higher values in the lower triangle, indicates that the model gives more weight to how recent states influence the interpretation of past information rather than vice versa. This sophisticated attention distribution mechanism enables our model to adaptively focus on the most relevant temporal information for each prediction task, contributing significantly to its accuracy in forecasting complex and dynamic traffic congestion patterns.

D. Ablation Study

To evaluate the contribution of each component in our proposed method, we conducted a comprehensive ablation study focusing on three key components: CL-DEC (Contrastive Learning-based Deep Embedded Clustering), TCN (Temporal Convolution Network), and DMAttention (Dependency Matrix Attention mechanism). We created six model variants: the full model (TCN-DMAttention with CL-DEC), a model without CL-DEC, a model without CL, a model without DEC (namely, CC and TCN-DMAttention), the TCN only, and the DMAttention only. TABLE VIII presents the results of our ablation study.

The results demonstrated the significant impact of each component: (1) Compared with the variant without CL-DEC, CL-DEC improved the performance of the entire model on all indicators, with accuracy weights increasing by 1.78%, F1w_weight increasing by 1.68%, AUC increasing by 1.55%, and accuracy increasing by 1.64%. This discovery indicates that CL-DEC has successfully extracted meaningful features from raw GPS data, enhancing the model's ability to distinguish different traffic congestion patterns. Among them, the two variants, a model without CL and a model without DEC, performed better than the variant without CL-DEC but inferior to the complete model. This finding indicates that both the CL and DEC components play important roles in the task of congestion state annotation. CL can effectively optimize feature representation, whereas DEC can alleviate the 'false positive problem' and enhance the discriminability and comparability of the model. (2) Only the TCN variant shows strong performance, but there is a gap with the complete model, highlighting the effectiveness of the TCN in capturing traffic data in terms of temporal dependence. The extended causal convolution in the TCN enables it to simulate long-term temporal relationships, which is crucial for predicting traffic congestion patterns that rely on recent and distant past events. (3) Although the DMAttention only variant is not as effective as the full model, this also indicates that the role of attention mechanism is important, and DMAttention has strong applicability in traffic congestion prediction tasks.

The full model's superior performance across all the metrics underscores the synergistic effect of combining CL-DEC, the TCN, and DMAttention. This combination allows the model to leverage robust feature representations, capture temporal dependencies, and focus on the most relevant features, thereby addressing the complex, time-varying nature of traffic congestion more effectively than any component in isolation. These results validate our integrated approach and highlight the importance of each component in tackling the challenges of traffic congestion prediction. The study also suggests potential avenues for future research, such as refining the interaction between these components or introducing additional complementary techniques to further enhance the model's performance.

VI. CONCLUSION

To overcome the dependence of existing methods for predicting traffic congestion via data annotation, we propose an end-to-end traffic congestion prediction framework that implements unsupervised learning. First, this paper simplifies the data structure of taxi GSP data via spatiotemporal rasterization to reduce the difficulty of analysis and then applies the improved CL-DEC algorithm to traffic data to complete the clustering of grid traffic congestion. Finally, the TCN-DMAttention model is constructed to predict short-term traffic congestion.

This paper analyses the interpretability of the clustering effect of the CL-DEC algorithm in terms of the average grid speed and average number of grid vehicle stops. On the basis of the clustering results, to analyze the prediction effect of the TCN-DMAttention model, in this paper, a total of 13 machine learning and deep learning models widely used in various fields are compared on four metrics: $Precision_{weighted}$, $F1_{weighted}$, AUC and $Accuracy$. Moreover, to analyze the robustness of the proposed model, in this paper, 20 random number seeds are taken to calculate the mean and variance of the prediction results. On the basis of the experimental results, the following conclusions are drawn. (1) The CL-DEC model incorporates multidimensional features into the scope of data annotation, combines discriminability and comparability, and results in accurate and interpretable clustering results. (2) The TCN structure can effectively represent the features of taxi GPS data after spatiotemporal rasterization. (3) The DMAttention structure can improve the feature extraction ability of the TCN. (4) The TCN-DMAttention model performs well

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on taxi GPS data, balances short-term fluctuations and long-term trends, and outperforms 13 mature comparative models in terms of performance. (5) The TCN-DMAttention model is robust and shows stable prediction results under the parameter initialization of 20 random number seeds.

In the future, more features can be mined for clustering and discretization of traffic data, and other relevant state features can be input into the prediction model to improve the practical application value of the framework proposed in this paper. On the other hand, we can further extend short-term forecasts to long-term forecasts by optimizing the model structure. In addition, relevant methods continue to be introduced to predict the passage time of vehicles in congested sections, which will help optimize the travel strategy and traffic resource allocation.

REFERENCES

[1] N. Zhong, J. Cao, and Y. Wang, "Traffic congestion, ambient air pollution, and health: Evidence from driving restrictions in Beijing," *J. Assoc. Environ. Resource Economists*, vol. 4, no. 3, pp. 821–856, Sep. 2017.

[2] V. Jain, A. Sharma, and L. Subramanian, "Road traffic congestion in the developing world," in *Proc. 2nd ACM Symp. Comput. Develop.*, Mar. 2012, pp. 1–10.

[3] W. Zheng, D.-H. Lee, and Q. Shi, "Short-term freeway traffic flow prediction: Bayesian combined neural network approach," *J. Transp. Eng.*, vol. 132, no. 2, pp. 114–121, Feb. 2006.

[4] B. Zhou, J. Liu, S. Cui, and Y. Zhao, "Large-scale traffic congestion prediction based on multimodal fusion and representation mapping," in *Proc. IEEE 9th Int. Conf. Data Sci. Adv. Anal. (DSAA)*, Oct. 2022, pp. 1–9.

[5] M. Akhtar and S. Moridpour, "A review of traffic congestion prediction using artificial intelligence," *J. Adv. Transp.*, vol. 2021, pp. 1–18, Jan. 2021.

[6] C. Song, Y. Lin, S. Guo, and H. Wan, "Spatial–temporal synchronous graph convolutional networks: A new framework for spatial–temporal network data forecasting," in *Proc. AAAI Conf. Artif. Intell.*, 2020, pp. 914–921.

[7] S. Wang and Z. Miao, "Anomaly detection in crowd scene," in *Proc. IEEE 10th Int. Conf. Signal Process.*, Oct. 2010, pp. 1220–1223.

[8] E. L. Manibardo, I. Laña, J. L. Lobo, and J. D. Ser, "New perspectives on the use of online learning for congestion level prediction over traffic data," in *Proc. Int. Joint Conf. Neural Netw.*, Jul. 2020, pp. 1–8.

[9] N. Kumar and M. Raubal, "Applications of deep learning in congestion detection, prediction and alleviation: A survey," *Transp. Res. C, Emerg. Technol.*, vol. 133, Dec. 2021, Art. no. 103432.

[10] M. Levin and Y.-D. Tsao, "On forecasting freeway occupancies and volumes (abridgment)," *Transp. Res. Rec.*, vols. 47–49, no. 773, 1980.

[11] X. Kong, Z. Xu, G. Shen, J. Wang, Q. Yang, and B. Zhang, "Urban traffic congestion estimation and prediction based on floating car trajectory data," *Future Gener. Comput. Syst.*, vol. 61, pp. 97–107, Aug. 2016.

[12] W. Cao and J. Wang, "Research on traffic flow congestion based on Mamdani fuzzy system," *AIP Conf. Proc.*, vol. 2073, no. 1, 2019, Art. no. 020101.

[13] M. G. Karlaftis and E. I. Vlahogianni, "Statistical methods versus neural networks in transportation research: Differences, similarities and some insights," *Transp. Res. C, Emerg. Technol.*, vol. 19, no. 3, pp. 387–399, Jun. 2011.

[14] Y. Li and C. Shahabi, "A brief overview of machine learning methods for short-term traffic forecasting and future directions," *SIGSPATIAL Special*, vol. 10, no. 1, pp. 3–9, Jun. 2018.

[15] X. Zhang, E. Onieva, A. Perallos, E. Osaba, and V. C. S. Lee, "Hierarchical fuzzy rule-based system optimized with genetic algorithms for short term traffic congestion prediction," *Transp. Res. C, Emerg. Technol.*, vol. 43, pp. 127–142, Jun. 2014.

[16] Y. Liu and H. Wu, "Prediction of road traffic congestion based on random forest," in *Proc. 10th Int. Symp. Comput. Intell. Design (ISCID)*, vol. 2, Dec. 2017, pp. 361–364.

[17] M. Fouladgar, M. Parchami, R. Elmasri, and A. Ghaderi, "Scalable deep traffic flow neural networks for urban traffic congestion prediction," in *Proc. Int. Joint Conf. Neural Netw. (IJCNN)*, May 2017, pp. 2251–2258.

[18] W. Zhang, Y. Yu, Y. Qi, F. Shu, and Y. Wang, "Short-term traffic flow prediction based on spatio-temporal analysis and CNN deep learning," *Transportmetrica A, Transp. Sci.*, vol. 15, no. 2, pp. 1688–1711, Nov. 2019.

[19] X. Di, Y. Xiao, C. Zhu, Y. Deng, Q. Zhao, and W. Rao, "Traffic congestion prediction by spatiotemporal propagation patterns," in *Proc. 20th IEEE Int. Conf. Mobile Data Manage. (MDM)*, Jun. 2019, pp. 298–303.

[20] Y.-Y. Chen, Y. Lv, Z. Li, and F.-Y. Wang, "Long short-term memory model for traffic congestion prediction with online open data," in *Proc. IEEE 19th Int. Conf. Intell. Transp. Syst. (ITSC)*, Nov. 2016, pp. 132–137.

[21] Z. Huang, J. Xia, F. Li, Z. Li, and Q. Li, "A peak traffic congestion prediction method based on bus driving time," *Entropy*, vol. 21, no. 7, p. 709, Jul. 2019.

[22] N. Ranjan, S. Bhandari, H. P. Zhao, H. Kim, and P. Khan, "City-wide traffic congestion prediction based on CNN, LSTM and transpose CNN," *IEEE Access*, vol. 8, pp. 81606–81620, 2020.

[23] S. Zhang, Y. Yao, J. Hu, Y. Zhao, S. Li, and J. Hu, "Deep autoencoder neural networks for short-term traffic congestion prediction of transportation networks," *Sensors*, vol. 19, no. 10, p. 2229, May 2019.

[24] M. Li and Z. Zhu, "Spatial–temporal fusion graph neural networks for traffic flow forecasting," in *Proc. 35th AAAI Conf. Artif. Intell.*, 2021, pp. 4189–4196.

[25] X. Kong, W. Xing, X. Wei, P. Bao, J. Zhang, and W. Lu, "STGAT: Spatial–temporal graph attention networks for traffic flow forecasting," *IEEE Access*, vol. 8, pp. 134363–134372, 2020.

[26] B. Lim, S. Ö. Arık, N. Loeff, and T. Pfister, "Temporal fusion transformers for interpretable multi-horizon time series forecasting," *Int. J. Forecasting*, vol. 37, no. 4, pp. 1748–1764, Oct. 2021.

[27] X. Chen, J. Wang, and K. Xie, "TrafficStream: A streaming traffic flow forecasting framework based on graph neural networks and continual learning," 2021, *arXiv:2106.06273*.

[28] J. Wang, Y. Mao, J. Li, Z. Xiong, and W.-X. Wang, "Predictability of road traffic and congestion in urban areas," *PLoS ONE*, vol. 10, no. 4, Apr. 2015, Art. no. e0121825.

[29] N. Chiabaut and R. Faitout, "Traffic congestion and travel time prediction based on historical congestion maps and identification of consensual days," *Transp. Res. C, Emerg. Technol.*, vol. 124, Mar. 2021, Art. no. 102920.

[30] M. Chen, X. Yu, and Y. Liu, "PCNN: Deep convolutional networks for short-term traffic congestion prediction," *IEEE Trans. Intell. Transp. Syst.*, vol. 19, no. 11, pp. 3550–3559, Nov. 2018.

[31] Q. Ren, D. Luo, and D. Song, "Rank supervised contrastive learning for time series classification," in *Proc. IEEE Int. Conf. Data Mining (ICDM)*, Dec. 2024, pp. 839–844.

[32] D. Luo et al., "Time series contrastive learning with information-aware augmentations," in *Proc. Conf. Artif. Intell. (AAAI)*, Jun. 2023, vol. 37, no. 4, pp. 4534–4542.

[33] E. Eldele et al., "Self-supervised contrastive representation learning for semi-supervised time-series classification," *IEEE Trans. Pattern Anal. Mach. Intell.*, vol. 45, no. 12, pp. 15604–15618, Dec. 2023.

[34] H. Cai, X. Zhang, and X. Liu, "Semi-supervised end-to-end contrastive learning for time series classification," 2023, *arXiv:2310.08848*.

[35] M. EskandariNasab, S. M. Hamdi, and S. F. Boubrahimi, "Enhancing multivariate time series-based solar flare prediction with multifaceted preprocessing and contrastive learning," in *Proc. Int. Conf. Mach. Learn. Appl. (ICMLA)*, Dec. 2024, pp. 369–376.

[36] Z. Wu, S. Pan, F. Chen, G. Long, C. Zhang, and P. S. Yu, "A comprehensive survey on graph neural networks," *IEEE Trans. Neural Netw. Learn. Syst.*, vol. 32, no. 1, pp. 4–24, Jan. 2021.

[37] X. Yin, G. Wu, J. Wei, Y. Shen, H. Qi, and B. Yin, "Deep learning on traffic prediction: Methods, analysis, and future directions," *IEEE Trans. Intell. Transp. Syst.*, vol. 23, no. 6, pp. 4927–4943, Jun. 2022.

[38] T. Gao, X. Yao, and D. Chen, "SimCSE: Simple contrastive learning of sentence embeddings," 2021, *arXiv:2104.08821*.

[39] T.-Y. Lin, P. Goyal, R. Girshick, K. He, and P. Dollár, "Focal loss for dense object detection," in *Proc. IEEE Int. Conf. Comput. Vis. (ICCV)*, Oct. 2017, pp. 2980–2988.

[40] J. Elman, "Finding structure in time," *Cognit. Sci.*, vol. 14, no. 2, pp. 179–211, Jun. 1990.

[41] S. Bai, J. Z. Kolter, and V. Koltun, "An empirical evaluation of generic convolutional and recurrent networks for sequence modeling," 2018, *arXiv:1803.01271*.

[42] Y. Zhang and L. Chen, "DDNet: A novel network for cerebral artery segmentation from MRA images," in *Proc. 12th Int. Congr. Image Signal Process., Biomed. Eng. Informat. (CISP-BMEI)*, Oct. 2019, pp. 1–5.

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[43] F. Rosenblatt, "The perceptron: A probabilistic model for information storage and organization in the brain," *Psychol. Rev.*, vol. 65, no. 6, pp. 386–408, 1958.
[44] R. Dey and F. M. Salem, "Gate-variants of gated recurrent unit (GRU) neural networks," in *Proc. IEEE 60th Int. Midwest Symp. Circuits Syst. (MWSCAS)*, Aug. 2017, pp. 1597–1600.
[45] A. Graves and A. Graves, "Long short-term memory," in *Supervised Sequence Labelling With Recurrent Neural Networks*, 2012, pp. 37–45.
[46] M. Beck et al., "XLSTM: Extended long short-term memory," in *Proc. Adv. Neural Inf. Process. Syst.*, 2024, pp. 107547–107603.
[47] V. Palo, "Decision tree methods: Applications for classification and prediction," *Shanghai Arch. Psychiatry*, vol. 27, no. 2, p. 30, 2015.
[48] T. Hastie, S. Rosset, J. Zhu, and H. Zou, "Multi-class AdaBoost," *Statist. Interface*, vol. 2, no. 3, pp. 349–360, 2009.
[49] M. P. LaValley, "Logistic regression," *Circulation*, vol. 117, no. 18, pp. 2395–2399, 2008.
[50] S. J. Rigatti, "Random forest," *J. Insurance Med.*, vol. 47, no. 1, pp. 31–39, 2017.
[51] G. Guo, H. Wang, D. Bell, Y. Bi, and K. Greer, "KNN model-based approach in classification," in *Proc. OTM Confederated Int. Conf., Move Meaningful Internet Syst.*Cham, Switzerland: Springer, Nov. 2003, pp. 986–996.
[52] C. Cortes and V. Vapnik, "Support-vector networks," *Mach. Learn.*, vol. 20, no. 3, pp. 273–297, Sep. 1995.

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